ModelIt!

Student Activity Pack

How Do Self-Driving Cars See?

Modeling LiDAR, Computer Vision, and Autonomous Decision-Making

Grade 11L3 | NGSS HS-PS4-1, HS-ETS1-2 | Systems and System Models

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **What do you already know about how self-driving cars detect objects around them, and what sensors do you think they use?**

2. **Why might a self-driving car have difficulty in heavy rain or fog when a human driver can still navigate safely?**

3. **Draw a diagram showing how you think a self-driving car builds a picture of its surroundings using different types of sensors.**

4. **What is the difference between a car that detects an obstacle and a car that decides what to do about it?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

LiDAR Resolution | Camera Detection Accuracy | Sensor Fusion Confidence | Processing Latency | Environmental Degradation | Decision Algorithm Reliability | Safety Margin

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Clear Highway Driving

Maximum LiDAR, minimal environmental degradation, increasing speed

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Heavy Rain Urban Intersection

40% LiDAR degradation, 30% camera degradation, complex urban scene

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Novel Object Emergency

Unclassified obstacle appears in the vehicle's path at highway speed

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

V2X Communication — Vehicle-to-everything communication allows autonomous vehicles to receive data from traffic signals, other vehicles, and road infrastructure — extending perception beyond line-of-sight sensors to include information about conditions around corners, over hills, and at distant intersections

Prediction Horizon — The time window over which the vehicle's algorithms predict the future positions of all detected moving objects — longer prediction horizons enable earlier and smoother responses but become exponentially less accurate as behavioral uncertainty compounds

Ethical Decision Framework — The programmed hierarchy of values the vehicle uses when a collision is unavoidable — how it prioritizes between protecting its passengers, protecting pedestrians, minimizing total harm, and following traffic laws when these objectives conflict

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

How does an autonomous vehicle build a real-time 3D understanding of its surroundings from raw sensor data — and can it ever be trusted to make life-or-death driving decisions? Use evidence from your simulation to support your answers.

1 How does your model demonstrate the fundamental trade-off between sensor resolution and processing latency?

2 Why is sensor fusion more valuable than simply using the best single sensor, and what happens when sensors provide contradictory information?

3 What does your model reveal about why autonomous vehicles perform differently in clear conditions versus adverse weather?

4 How would your model change if processing latency could be reduced to near-zero through advanced computing hardware?

5 What are the limitations of modeling autonomous vehicle safety when real-world driving includes situations that have never occurred before?

8 STEM Challenge: Design an Autonomous Vehicle Sensor Suite

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YOUR ENGINEERING MISSION

Design an optimal sensor configuration for an autonomous vehicle that must operate safely in your city's specific driving conditions, addressing sensor selection, placement, fusion architecture, and failure mode handling.

THE SCENARIO

A transportation company is deploying autonomous delivery vehicles in your city and has hired your team to design the sensor suite and perception system. Your city has specific challenges — weather patterns, road conditions, pedestrian density, and infrastructure — that the sensor system must handle. You must select and position sensors, design the fusion architecture, establish minimum safety margins for different conditions, and create protocols for what the vehicle does when sensor confidence drops below safe thresholds.

Engineering Design Thinking

1. HOW MANY AND WHAT TYPE OF LIDAR UNITS WOULD YOU MOUNT ON THE VEHICLE AND WHERE WOULD YOU POSITION THEM TO MINIMIZE BLIND SPOTS?

2. WHAT CAMERA SPECIFICATIONS (RESOLUTION, FRAME RATE, FIELD OF VIEW, LOW-LIGHT SENSITIVITY) ARE REQUIRED FOR RELIABLE OBJECT CLASSIFICATION AT HIGHWAY SPEEDS?

3. WHAT IS YOUR MAXIMUM ACCEPTABLE PROCESSING LATENCY AND HOW DOES THIS CONSTRAIN YOUR COMPUTING HARDWARE REQUIREMENTS?

4. WHAT FAILURE CASCADE PROTOCOL WOULD YOU IMPLEMENT WHEN ENVIRONMENTAL CONDITIONS DEGRADE SENSOR CONFIDENCE BELOW YOUR SAFETY THRESHOLD?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Autonomous Vehicle Engineers design and test self-driving systems at companies like Waymo, Cruise, and Aurora, integrating sensors, perception algorithms, and decision-making software, earning $120,000-$220,000/year. Robotics Perception Scientists develop the computer vision and sensor fusion algorithms that allow machines to understand their environment, earning $110,000-$200,000/year. Transportation Safety Analysts evaluate autonomous vehicle performance data and develop safety standards for regulatory agencies, earning $80,000-$140,000/year.